

TIC 348762141

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 17.938393 d (ALIASED, see below)
-> corrected period: 35.876786 d
Epoch (t0) 1342.7463 BTJD
Duration 0.1700 d
Depth 12.092 % (120,921 ppm)
SNR 571.4

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

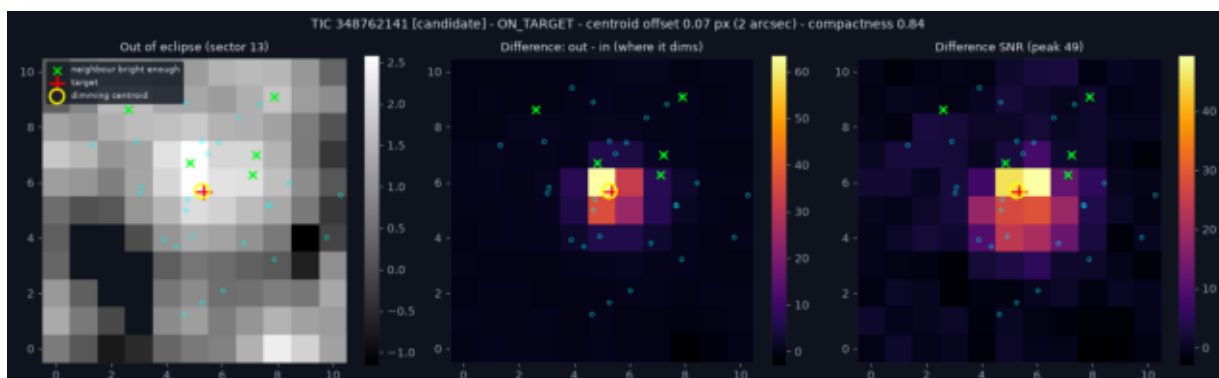
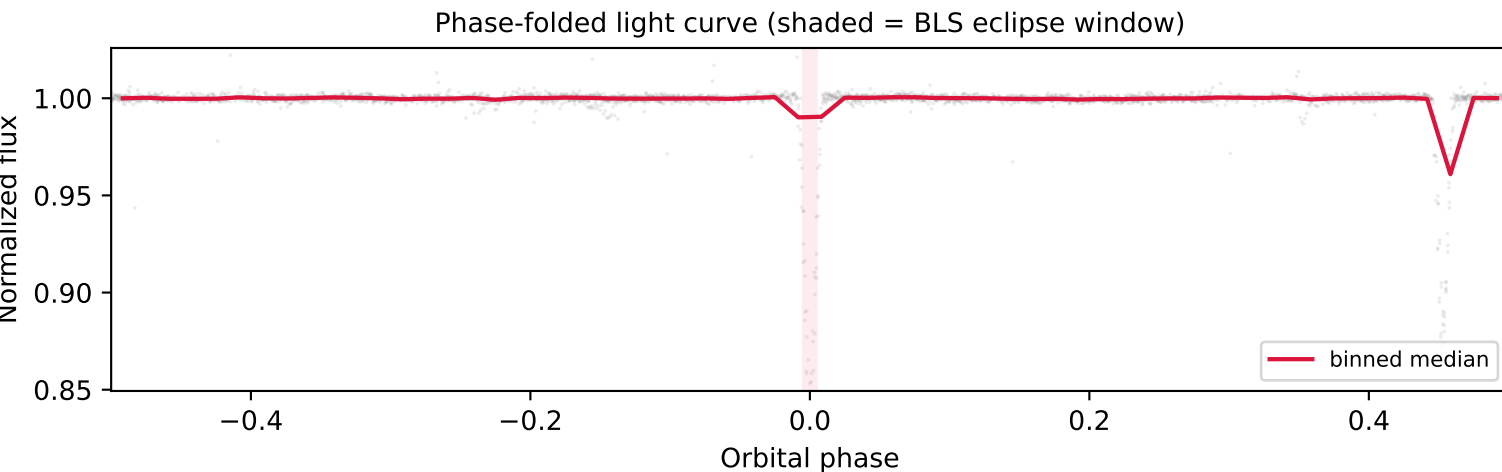
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.02 (gate ≥ 0.50)
Robust depth sigma 431.9 (gate ≥ 7.0)
Odd/even z-score 5.99 (ALIAS FLAGGED)
Secondary eclipse detected, phase 0.46, 9.9 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 13 light-curve dip in this sector: 207.1 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 49.42 (gate ≥ 5.0) Compactness 0.84 (gate ≥ 0.50)
Dimming centroid offset 0.07 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 348762143
1.10 px away, dTmag -0.90 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 5.99 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 35.876786 d, in any write-up.